ARAVINDAN RAMESHBABU

SENIOR DATA & AI ENGINEER

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PROFESSIONAL SUMMARY

Senior Data & AI Engineer with **5+ years** of expertise architecting high-throughput cloud data platforms, distributed lakehouses, and applied GenAI workflows. Proven engineering leadership directing a 5-member team on EKS-managed Apache Iceberg/Spark platforms; deep expertise in multi-tenant ETL orchestration, SCD Type 1 & 2 history pipelines, Databricks Auto Loader, Delta Live Tables (DLT), and Model Context Protocol (MCP) integrations for autonomous data management.

TECHNICAL EXPERTISE

Big Data & Lakehouse:	Apache Spark (PySpark, Spark SQL, Spark Streaming), Apache Iceberg, Delta Lake, Trino, Nessie Catalog, Hive Metastore
Streaming & Ingestion:	Apache Kafka, Spark Structured Streaming, Databricks Auto Loader, Change Data Feed (CDF), Declarative CDC (SCD 1 & 2)
Cloud & Infrastructure:	AWS (EKS, S3, Glue, Lambda, EC2, IAM / IRSA), Databricks, GCP, Kubernetes (K8s), Docker, GitHub Actions, Linux / Bash
AI & GenAI Platforms:	Model Context Protocol (MCP), RAG Architectures, ChromaDB, Ollama, Open-Source LLMs, SeqGAN Synthetic Generation
Orchestration & DBs:	Apache Airflow, PostgreSQL, MySQL, MongoDB, Cassandra, Dremio, Unity Catalog
Languages & Core:	Python, SQL, PySpark, Scala, Multi-Tenant Architecture, Medallion Architecture, CI/CD

PROFESSIONAL EXPERIENCE

Senior Data Engineer (SDE 2) · Engineering Team Lead
Feb 2025 – Present

M2P Fintech · Chennai, India

- **EKS Open-Source Lakehouse:** Architected and deployed an open-source data lakehouse on AWS EKS integrating Apache Spark, Apache Iceberg, Trino, and Nessie Catalog, slashing data access latency and cutting storage overhead.
- **Agentic MCP Integration:** Pioneered an agentic data stack driving ETL workflows via Model Context Protocol (MCP), empowering LLM agents to trigger, query, and monitor data pipelines via a conversational control plane.
- **Fintech SCD-2 Audit Trails:** Engineered SCD Type 2 history pipelines on Apache Iceberg with dynamic watermarking and Hot/Cold table partitioning, establishing zero-loss fintech audit compliance and rolling KPI aggregations.
- **Serverless ETL Automation:** Automated serverless pipelines using AWS Glue and Lambda, reducing manual schema maintenance by **40%** and executing automated real-time schema validation across tenant streams.
- **Enterprise Security & Monitoring:** Hardened distributed workloads using IRSA (IAM Roles for Service Accounts) for Spark and Trino; integrated automated webhook alerting for immediate pipeline failure mitigation.
- **Leadership & Accolades:** Led a 5-member engineering team to deliver the end-to-end lakehouse platform ahead of schedule; honored with the **Above & Beyond in Engineering (Applause 2026)** award for technical ownership.

Data Engineer
Apr 2021 – Feb 2025

Tata Consultancy Services · Chennai, India

- **Kubernetes Spark Optimization:** Developed and tuned large-scale PySpark data processing jobs on Kubernetes-managed Spark clusters for telecom advanced analytics.
- **Airflow Orchestration:** Automated fault-tolerant ETL workflows with Apache Airflow DAGs on Kubernetes, elevating compute resource utilization and pipeline stability.
- **Enterprise Cloud Migration:** Led the migration of legacy Oracle RDBMS to GCP Cloud leveraging PySpark and Spark-SQL, accelerating job completion times and eliminating data consistency issues.
- **Storage & Query Tuning:** Optimized storage by implementing partitioned Hive tables with compression across AWS S3 and GCP, reducing storage footprint and read latency.
- **Real-Time Kafka Streaming:** Integrated Spark Streaming and Apache Kafka for high-throughput telecom event consumption; migrated VDS to Dremio for optimized federated data access.
- **Recognition:** Awarded **Technical Excellence (Aug 2022)** and **Inspiring Talent (Jun 2023)** for outstanding project delivery and technical innovation across data initiatives.

FEATURED ENGINEERING PROJECTS

Goodcabs — Transportation Lakehouse & Declarative CDC Pipeline
Personal Engineering Project · 2025

Databricks · Spark Declarative Pipelines · Auto Loader · Delta Lake · SCD Type 1 CDC

- **Medallion Lakehouse on Databricks:** Architected an end-to-end Bronze-Silver-Gold pipeline using Spark Declarative Pipelines (DLT) processing full and incremental cab transactions across 10 Indian Tier-2 cities.
- **Auto Loader Streaming Ingestion:** Deployed Databricks Auto Loader (`cloudFiles`) with automated column type inference and rescue schema evolution mode, eliminating pipeline stops from upstream schema drift.
- **Declarative CDC Flow:** Engineered automated SCD Type 1 Change Data Capture upserts via `dp.create_auto_cdc_flow` with `sequence_by` timestamp ordering to automatically resolve out-of-order records.
- **Quality Gates & Dimensions:** Enforced data quality expectations (`@dp.expect`), dynamic calendar dimension generation with Indian national holiday flags, and pre-filtered Gold regional marts.

Prism — Agentic Control Plane for Data Lakehouse
M2P Fintech · 2025

Apache Airflow · Apache Spark · Trino · Unity Catalog · Model Context Protocol · Iceberg

- **Control Plane Architecture:** Built a unified platform control plane hosting Airflow, Spark, Trino, and Unity Catalog to orchestrate ingestion from multi-source databases into Iceberg and Delta formats.
- **MCP-Driven Natural Language ETL:** Exposed source/destination connections, schema migrations, and pipeline triggers as Model Context Protocol tools, enabling chat-driven pipeline orchestration for engineering teams.

Synthetic Telecom Data & RAG/MCP Orchestration Platform
Tata Consultancy Services · 2023 – 2024

SeqGAN · RAG Architectures · ChromaDB · Ollama · Apache Airflow · Python

- **Generative Synthetic Data:** Developed a SeqGAN deep learning architecture generating high-fidelity synthetic telecom datasets to overcome model training bias on imbalanced customer records.
- **Enterprise RAG Retrieval:** Built domain-specific semantic search using ChromaDB vector embeddings and Ollama-hosted LLMs, orchestrated end-to-end via Apache Airflow with MCP agent tool bindings.

Real-Time Location-Based Marketing Stream
Tata Consultancy Services · 2022

Spark Streaming · Apache Kafka · Cassandra NoSQL · GDPR Compliance

- **Real-Time Spatial Stream:** Ingested high-velocity ISP tower location events with Spark Streaming, cross-referencing user profiles to generate sub-second geo-targeted promotions.
- **GDPR Audited Storage:** Designed a Cassandra NoSQL sliding-window store with automated 6-month TTL cleanup, delivering compliant real-time feeds to external partner APIs.

EDUCATION

B.E. Electronics & Communication — **2016 – 2020**

University College of Engineering, Anna Univ

- First Class (CGPA 7.05). Final-year OpenCV Face Recognition graded 'S' (Top Grade).
- Elected Chairman of Student Cell; spearheaded technical symposiums.

HONORS & AWARDS

Above & Beyond in Engineering – M2P Fintech — **Aug 2026**

Inspiring Talent Award – TCS — **Jun 2023**

Technical Excellence Award – TCS — **Aug 2022**

CERTIFICATIONS & CREDENTIALS

Data Streaming Engineer — Apache Flink — Confluent · 2026

Azure Data Engineer Associate (DP-203) — Microsoft

AWS Solutions Architect — Associate — Amazon Web Services

Linux Essentials (LE-01) — Linux Professional Inst.

AI for Real-World Applications — Specialization